Persistent Branches and Transcritical Exchange

Thursday, September 17, 2026

Definition: Persistent equilibrium branch

Let $R \subseteq \mathbb{R}$ be a specified parameter set. For $\dot{x} = f(x; r)$, a **persistent equilibrium branch on** R is a continuous function $x^* : R \to \mathbb{R}$ satisfying

$$f(x^*(r); r) = 0 \qquad \text{for every } r \in R.$$

Definition: Transcritical bifurcation

A **transcritical bifurcation** occurs when two equilibrium branches cross and exchange stability.

Definition: Transcritical normal form

The **transcritical normal form** is

$$\dot{x} = rx - x^2.$$

Question: How does the transcritical normal form exchange stability?

Analyze

$$\dot{x} = rx - x^2 = x(r - x).$$

Solution. The equilibrium branches are

$$x_0^*(r) = 0, \qquad x_1^*(r) = r.$$

Since

$$f_x(x; r) = r - 2x,$$

the branch rates are

$$\lambda_0(r) = f_x(0; r) = r, \qquad \lambda_1(r) = f_x(r; r) = -r.$$

Therefore

	$x^* = 0$	$x^* = r$
$r < 0$	asymptotically stable	unstable
$r > 0$	unstable	asymptotically stable

At $r = 0$, $\dot{x} = -x^2 \le 0$, so the origin repels from the left and attracts from the right; it is half-stable.

Visualization: Transcritical branch crossing

r	ordered equilibria	phase line
$r<0$	$r<0$	$\longleftarrow\ \circ_{x=r}\ \longrightarrow\ \bullet_{x=0}\ \longleftarrow$
$r=0$	0	$\longleftarrow\ \odot_{x=0}\ \longleftarrow$
$r>0$	$0<r$	$\longleftarrow\ \circ_{x=0}\ \longrightarrow\ \bullet_{x=r}\ \longleftarrow$

The bifurcation diagram is the crossing of

$$x^*=0 \qquad \text{and} \qquad x^*=r,$$

with solid and dashed segments exchanged at $(r,x^*)=(0,0)$.

Question: Where is the bifurcation curve in a two-parameter model?

For

$$\dot{x}=x(1-x^2)-a(1-e^{-bx}),$$

find the transcritical bifurcation curve at $x=0$ and the leading nonzero equilibrium branch.

Solution. The origin is an equilibrium for every (a,b). Fix a point (a_0,b_0) satisfying $a_0b_0=1$ and $b_0\neq 0$. All remainder estimates in the next two displays are as $x\to 0$, uniformly for (a,b) in a sufficiently small fixed neighborhood of (a_0,b_0). Since

$$1-e^{-bx}=bx-\frac{1}{2}b^2x^2+O(x^3),$$

we have

$$\dot{x}=(1-ab)x+\frac{1}{2}ab^2x^2+O(x^3).$$

The linear coefficient vanishes on

$$\boxed{ab=1}.$$

For parameters near (a_0,b_0) and $x^*\neq 0$, division by x^* gives

$$0=1-ab+\frac{1}{2}ab^2x^*+O((x^*)^2),$$

so

$$x^*=\frac{2(ab-1)}{ab^2}+O((ab-1)^2) \qquad ((a,b)\to(a_0,b_0)).$$

The leading branch rates are

$$\lambda_0=1-ab, \qquad \lambda_1=ab-1+O((ab-1)^2) \qquad ((a,b)\to(a_0,b_0)),$$

and therefore have opposite signs for all sufficiently nearby parameters with $ab\neq 1$.

Question: How is a nonzero persistent branch moved to the origin?

Analyze

$$\dot{x} = r \log x + x - 1, \qquad x > 0,$$

near the persistent equilibrium $x = 1$.

Solution. Set

$$u = x - 1.$$

All remainder estimates below are as $u \to 0$, uniformly for r in a fixed neighborhood of -1. Using

$$\log(1+u) = u - \frac{1}{2}u^2 + O(u^3),$$

we obtain

$$\dot{u} = (r+1)u - \frac{r}{2}u^2 + O(u^3).$$

The bifurcation value is

$$r_c = -1.$$

For r near -1, define

$$R = r + 1, \qquad X = \frac{r}{2}u = \frac{r}{2}(x-1).$$

Because r is constant in time,

$$\dot{X} = RX - X^2 + O(X^3) \qquad (X \to 0).$$

Thus the local equation has transcritical normal form to quadratic order.

Lemma: Factorization forced by a persistent zero branch

Let $f \in C^2$ near $(0,0)$ and suppose $f(0;r) = 0$ for all sufficiently small r. Then

$$f(x;r) = xg(x;r)$$

for a C^1 function g.

Proof. Define

$$g(x;r) = \int_0^1 f_x(sx;r)\,ds.$$

Then

$$xg(x;r) = \int_0^x f_x(u;r)\,du = f(x;r) - f(0;r) = f(x;r).$$

□

Proposition: Exchange of stability on two local branches

For the parameterized system $\dot{x} = f(x;r)$, let $f(x;r) = xg(x;r)$, where $g \in C^1$ near $(0,0)$, and suppose

$$g(0;0) = 0, \qquad a = g_r(0;0) \neq 0, \qquad b = g_x(0;0) \neq 0.$$

Then the local equilibrium branches and their linearized rates are

$$x_0(r) = 0, \qquad x_1(r) = -\frac{a}{b}r + o(r) \qquad (r \to 0),$$

$$\lambda_0(r) = ar + o(r), \qquad \lambda_1(r) = -ar + o(r) \qquad (r \to 0).$$

Hence the branches exchange stability at $r = 0$.

Proof. Since

$$g(x;r) = ar + bx + o(|r| + |x|) \qquad ((x,r) \to (0,0)),$$

the implicit function theorem gives

$$g(x_1(r);r) = 0, \qquad x_1(r) = -\frac{a}{b}r + o(r) \qquad (r \to 0).$$

On $x_0(r) = 0$,

$$\lambda_0(r) = f_x(0;r) = g(0;r) = ar + o(r) \qquad (r \to 0).$$

On the nonzero branch, $g(x_1(r);r) = 0$, so

$$\lambda_1(r) = x_1(r)g_x(x_1(r);r) = -ar + o(r) \qquad (r \to 0).$$

The signs are opposite for all sufficiently small $r \neq 0$. $\square$

Question: Is a rational vector field still transcritical?

Analyze

$$\dot{x} = \frac{x(r+x)}{1+x^2}.$$

Solution. The equilibrium branches are

$$x_0^* = 0, \qquad x_1^* = -r.$$

At either equilibrium, the numerator vanishes, so differentiation gives

$$f_x(x^*;r) = \frac{r + 2x^*}{1 + (x^*)^2}.$$

Therefore

$$f_x(0;r) = r, \qquad f_x(-r;r) = -\frac{r}{1+r^2}.$$

The rates have opposite signs for $r \neq 0$. At $r = 0$,

$$\dot{x} = \frac{x^2}{1+x^2} \geq 0,$$

so the origin is half-stable. Near $(x,r) = (0,0)$,

$$\dot{x} = rx + x^2 + O(|x|^3 + |r|x^2) \qquad ((x,r) \to (0,0)).$$

Question: Is a rational vector field still transcritical? (continued)

The state reversal $y = -x$ gives the leading form

$$\dot{y} = ry - y^2.$$

Idea: Mathematical and physical branches

If x denotes a population, concentration, or intensity, the physical state space may be $x \geq 0$. A branch with $x^* < 0$ remains part of the mathematical bifurcation diagram but is not a physically admissible state.